

Local E-Government Toolkit

Empowering Local Governments Through Digital Transformation

Technology



6. Technology

Indicator	Description
1. #214 Contact details	Evidence of a “contact us” feature (e.g. phone numbers, email addresses, physical addresses, and sometimes online forms)
2. #201 Browser compatibility	The ability of the portal to function correctly and consistently across different web browsers, operating systems, and devices.
3. #202 Ease of portal finding	Ease of finding the municipal government portal on the first result page of the three most popular search engines used in the country.
4. #204 Mobile device accessibility	The portal displays and works well (is navigable and usable) when accessed through a mobile device (e.g., smartphone, tablet).
5. #205 Navigability	A sitemap or index with a list of all the available pages within the website accessible to users.
6. #207 Internal advanced search mechanism	Advanced search options that allow users to run custom searches with specific text, indicate unwanted words, specify date ranges, language options, etc.



6. Technology (cont.)

Indicator	Description
7. #208 Alignment with markup validation standards	Compliant with the markup validity standards set by the World Wide Web Consortium (W3C).
8. #209 Alignment with display standards	Compliant with the CSS style sheet standards set by the World Wide Web Consortium (W3C).
9. #210 Alignment with accessibility standards	Compliant with the WCAG 2.1 LEVEL AA standards set by the W3C.
10. #246 Personal/ Business data accessibility	The ability for users to access their own data (local government has on record).
11. #247 Personal/ Business data updating	The ability for users to modify their own data (local governments have on record).
12. #288 Helpdesk call number	A helpdesk call number for supporting people who face difficulties in executing their services online.

1. Contact details (#214)

What	Commonly known as a “contact us” feature, it can come in a variety of formats (e.g. phone numbers, email addresses, physical addresses, and sometimes online forms)
Why	Enables easy and direct communication between residents, businesses, and government officials, fostering transparency, accessibility, and efficient resolution of civic issues and inquiries.
How?	Indicative steps: 1. Create a Dedicated Page: a. Design a clear and user-friendly webpage labeled "Contact Us" or "Get in Touch" Regular Updates and Maintenance: 2. Provide Comprehensive Information: a. Include phone numbers, email addresses, and physical addresses of government offices. 3. Include Operating Hours: a. Specify office hours and days of operation for each department to manage visitor expectations. 4. Prominent Placement: a. Feature the contact details prominently on the website, preferably in the header, footer, or a dedicated sidebar, ensuring high visibility. 5. Regular Updates: a. Keep contact information up-to-date, For more information check out Nielson Norman Group guide or US EPA guide.
Case Examples	Check out Colombo's Contact Us Page. 

Submit your case [here](#)



2. Browser compatibility (#201)

What	Refers to the ability of the portal to function correctly and consistently across different web browsers, operating systems, and devices. By accessing http://gs.statcounter.com/browser-market-share and checking for the three most popular web browsers used in the country, the municipality website's compatibility can be assessed by checking on each of these three web browsers.
Why	Ensures that all residents, regardless of their choice of web browser or device, can access essential information and services easily, promoting inclusivity and enhancing civic engagement.
How?	Indicative steps: 1. Identify Target Languages a. Determine which languages are widely spoken within the community or region served by the municipal government. 2. Content Management System (CMS): a. Choose a CMS that supports multilingual content. 3. Regular Updates and Maintenance: a. Continuously update translated content to reflect changes in policies, services, or important announcements. For more information check out <i>the</i> UK Gov's guide or W3C's guide
Case Examples	Check out https://gs.statcounter.com/search-engine-market-share to see your region's main browser share



3. Ease of portal finding (#202)

What	Ease of finding the municipal government portal is understood as being found on the first result page of the three most popular search engines used in the country, this can be checked by accessing http://gs.statcounter.com/browser-market-share to see which are the most popular search engines
Why	Most users rarely venture beyond the first page, this ensures easy access to vital information and services, enhancing user satisfaction, and fostering efficient civic engagement.
How?	Indicative steps: 1. Optimize Website Content: a. Use relevant keywords and phrases related to municipal services and location throughout the website content to enhance search engine optimization (SEO) 2. Mobile-Friendly Design: a. Ensure the website is responsive and mobile-friendly, as search engines prioritize mobile-friendly sites 3. Local SEO: a. Utilize local SEO techniques, including adding the municipality to meta titles, meta descriptions, and headers, to enhance visibility in local searches. 4. Regular Content Updates: a. Keep the website content fresh and up-to-date, as search engines prefer websites that provide timely and relevant information. For the complete guide by the current author click here.
Case Examples	For more information check out <i>the</i> Semrush Blog : or Google's Developer guide



4. Mobile device accessibility (#204)

What	A responsive web design when the portal displays and works well (is navigable and usable) when accessed through a mobile device (e.g., smartphone, tablet). The portal could be provided through a regular web browser (responsive/adaptive website) or through a mobile application.
Why	Ensures residents can conveniently access vital services, and information, and engage with local government, fostering inclusivity and enhancing civic participation on smartphones and tablets.
How?	Indicative steps: 1. Responsive Web Design: a. Implement a responsive web design adapting the layout and content based on the user's device 2. Mobile-Friendly Navigation: a. Simplify navigation for mobile users by using clear menus, intuitive icons, and a streamlined menu structure. 3. Readable Text and Images: a. Use legible font sizes and ensure images are appropriately sized for mobile screens. 4. Touch-Friendly Buttons: a. Design interactive elements, such as buttons and links, to be touch-friendly, ensuring they are large enough and well-spaced 5. Mobile Compatibility Testing: a. Regularly test the portal on different mobile devices and browsers to identify and address any issues related to display, functionality, or user experience.
Case Examples	For more information check out <i>the</i> W3C's Mobile Accessibility Guide or the US GSA's guide .





5. Navigability (#205)

What	Navigability is checked via a sitemap or index that has a list of pages of a web site accessible to users. It is a web page that lists all the available pages within the website, typically organized in a hierarchical fashion. This helps visitors and search engine bots find pages on the site.
Why	Crucial as it provides a structured, organized map of the website's content, making it easier for visitors and search engines to navigate and access relevant information efficiently.
How?	Indicative steps: 1. Choose the Right Format: a. Decide between an XML sitemap (for search engines) and an HTML sitemap (for users). XML sitemaps are essential for search engines, while HTML sitemaps serve as user-friendly guides. 2. Link from Website Footer: a. Add a link to the HTML sitemap in the footer section of your website. 3. Submit XML Sitemap to Search Engines: a. Submit the XML sitemap to search engines like Google and Bing through their respective webmaster tools. This helps search engines crawl and index your site more effectively. 4. Update Sitemap Regularly: a. Keep the XML sitemap updated whenever you add new pages For more information check out Google's guide.
Case Examples	Check out Casablanca's Sitemap (in French).





6. Internal advanced search mechanism (#207)

What	Advanced search options may include features that allow users to run custom searches with specific text, indicate unwanted words, specify date ranges, language options, etc. There can also be options available to refine the results after the search page.
Why	It allows users to refine their queries, enabling precise and efficient access to specific information or services, enhancing user satisfaction and usability.
How?	Indicative steps: 1. Understand User Needs: a. Identify the specific search requirements of your users, 2. Select a Robust Search Engine: a. Choose a powerful search engine or software that supports advanced search functionalities 3. Faceted Search and Filters: a. Implement faceted search, allowing users to filter results by categories like departments, document types, dates, or other relevant attributes 4. Natural Language Processing (NLP): a. If feasible, integrate NLP technology to enable users to input queries in natural language 5. Continuous Improvement: a. Regularly assess user feedback and search analytics to identify areas for enhancement. For more information check out UXPin's guide or HubSpot Tips
Case Examples	Check out Riyadh 's advanced search options





7. Alignment with markup validation standards (#208)

What	Defined as compliant with the markup validity standards set by the World Wide Web Consortium (W3C). This can be verified via https://validator.w3.org/ inserting the link of the municipality, using the default options and clicking ""check"". Evidence of "read timeout" or "errors" by the tool indicates a lack of alignment.
Why	Ensures accessibility, interoperability, and a consistent user experience, promoting inclusivity and adherence to global web development best practices.
How?	Indicative steps: 1. Use Valid HTML/CSS: a. Write well-structured, semantic HTML and CSS code. Validate the code using W3C's validation services 2. Responsive Design: a. Implement responsive design to ensure the portal is accessible and functions on all devices. 3. Alt Text for Images: a. Include descriptive alt text for images, enabling users with visual impairments to understand the content of the images through screen readers. 4. Color Contrast: a. Maintain sufficient color contrast between text and background colors to ensure readability. 5. Keyboard Accessibility: a. Ensure all interactive elements, such as links and buttons, are navigable and usable with a keyboard. 6. User Testing: a. Involve users with disabilities in usability testing to gain insights into their experiences.
Guide	For more information check out W3C's guide .

 United Nations	Department of Economic and Social Affairs 8. Alignment with display standards (#209) 
What	Defined as compliant with the CSS style sheet standards set by the World Wide Web Consortium (W3C). Can be checked via https://jigsaw.w3.org/css-validator/ and by pasting the link of the municipality using the default options and clicking "check. Evidence of "errors" by the tool indicates a lack of alignment.
Why	Ensures consistent design, efficient maintenance, and seamless user experience across various devices, promoting accessibility and enhancing the site's credibility and professionalism.
How?	Indicative steps: 1. Use External Style Sheets: a. Utilize external CSS files rather than inline styles. 2. Valid CSS Code: a. Write clean and valid CSS code. Validate your stylesheets using W3C's CSS Validation Service 3. Responsive Design: a. Implement responsive design techniques using media queries to ensure the website layout adjusts appropriately on different devices and screen sizes. 4. Semantic HTML: a. Use semantic HTML elements and apply CSS styles appropriately. 5. Testing Across Browsers: a. Regularly test the website's styles across various web browsers and devices. 6. Regular Updates: a. Stay updated with the latest CSS standards and best practices.
Guide	For more information check out W3C's guide .

 United Nations	Department of Economic and Social Affairs	9. Alignment with Accessibility standards (#210)  
What	Defined as compliant with the Web Content Accessibility Guidelines (WCAG2.1), specifically the WCAG 2.1 LEVEL AA standards set by the World Wide Web Consortium (W3C). This can be verified via https://www.tawdis.net/ with default Level AA options selected and pasting the URL link in the box and pressing analyzing. Evidence of "problems" indicated by the summary page.	
Why	Ensures accessibility for all users, including those with disabilities, promoting inclusivity, equal access to information, and a seamless online experience for everyone.	
How?	Indicative steps: 1. Accessible Content: - Ensure all website content, including text, images, videos, and documents, is perceivable, operable, understandable, and robust for all users 2. Keyboard Accessibility: - Ensure all interactive elements and functionalities can be accessed and used with a keyboard alone 3. Focus Styles: - Implement visible and clear focus styles for interactive elements to indicate focus and active states, aiding keyboard users. 4. Regular Audits and Updates: - Conduct regular accessibility audits, addressing any issues promptly. For more information check out W3C's guide.	
Case Examples 	Check out Santo Domingo's Accessibility widget on the left-hand side.	



10. Personal/ Business Data accessibility (#246)

What	Defined as the ability for users to access their own data (that local government has on record), on the government portal, such as (but not limited to) data on social security, vehicle registration, etc.
Why	Ensures transparency, fosters public trust, and empowers people to access their own information, promoting accountability and civic engagement.
How?	Indicative steps: 1. Review Privacy Laws: - Understand laws governing the accessibility and protection of personal/Business data. 2. Implement Secure Authentication: - Require strong, multi-factor authentication for users accessing personal/business data. Use encrypted connections (SSL/TLS) to protect data in transit. 3. User Verification: - Verify the identity of users before granting access to personal data. 4. Data Minimization: - Only collect and display essential personal data necessary for municipal services. 5. Secure Storage: - Store personal data in encrypted databases with regular security updates and patches. 6. Legal Disclaimer: - Clearly state the terms of use and privacy policies on the website.
Case Examples	For more Warsaw's page on personal data processing or Check out Singapore's business tax registration app.





11. Personal/ Business Data updating (#247)

What	Defined as the ability for individuals to modify part of their own data (that local governments have on record).
Why	Vital for people to maintain accurate information, ensuring efficient communication, improved service delivery, and fostering a sense of control and ownership over their data.
How?	Indicative steps: User Account Creation - Registration Process: Design a simple registration form requiring essential details. - Email Verification: Include an email verification step to authenticate user accounts. Accessing Personal Profiles - Provide a well-organized dashboard upon login, displaying categories like 'Personal Details.' or 'Business Information' - Create an intuitive interface enabling easy navigation to personal data sections. Modifying Data - Data Categories: Categorize data into clear sections. - Edit Functionality: Implement an 'Edit' button within each section, allowing users to modify relevant data. - Data Validation: Ensure validation checks for accurate and complete information entry. - Real-time Updates: Enable real-time saving of modifications to prevent data loss. For more information check out <i>the current author's guide</i>.
Case Examples	For more Tallinn's informative page on personal data updating or Check out Dubai's Smart Supplier App allowing businesses to register and modify their data





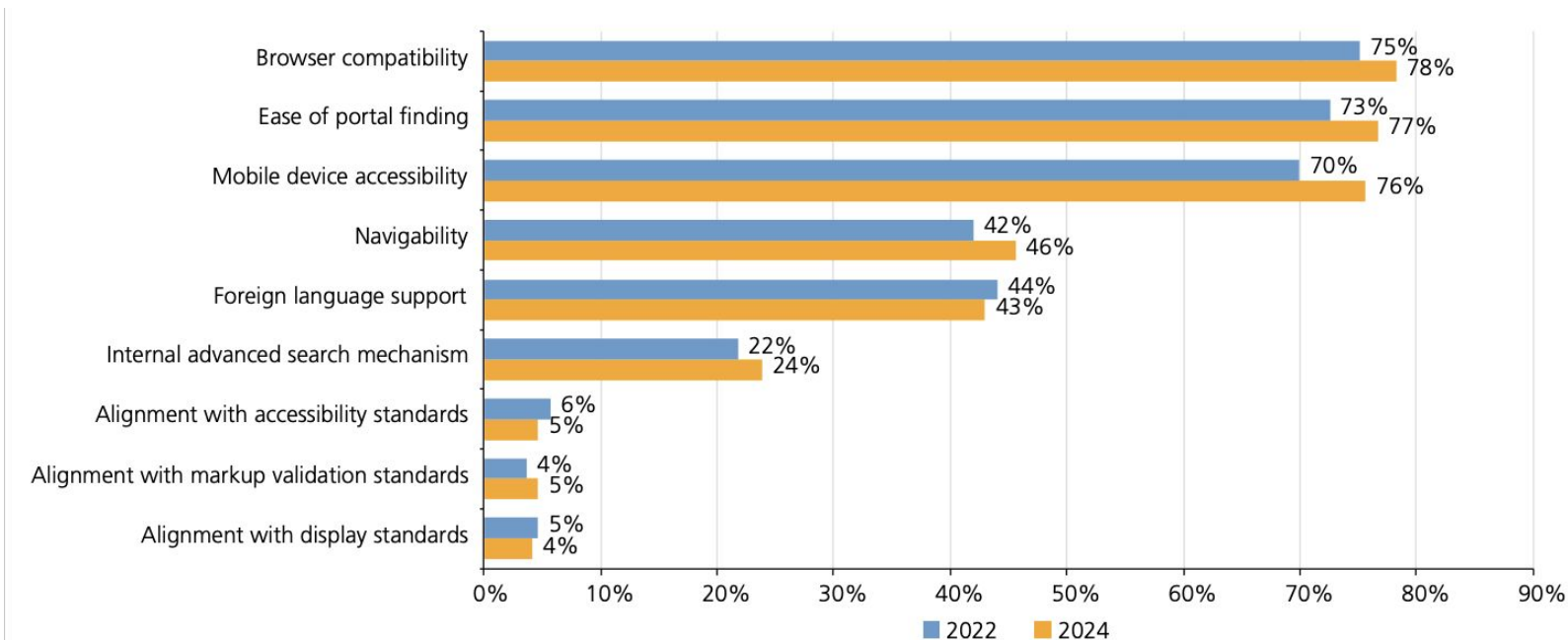
12. Helpdesk call number (#288)

What	Defined as a helpdesk call number for supporting people who face difficulties in executing their services online or through any digital channel.
Why	Provides efficient and accessible support to residents and businesses, ensuring timely assistance, and enhancing overall community satisfaction with government services.
How?	Indicative steps: 1. Choose a Dedicated Number: a. Select a local or toll-free phone number specifically for the helpdesk. 2. Website Placement: a. Clearly display the helpdesk number on the website, preferably in the header or footer, ensuring it's visible on every page. 3. Interactive Voice Response (IVR): a. Implement an IVR system to guide callers to the right department based on their query. 4. Feedback Mechanism: a. Integrate a feedback system to gauge user satisfaction and identify areas for improvement in the helpdesk services. 5. Regular Updates: a. Keep the contact information, including the helpdesk number.
Case Examples	Check out Istanbul's ALO153 helpdesk service





Implementation of technology indicators in city portals (percentage of cities)



Source: UN E-Government Survey 2024, UN DESA

These numbers are represented by the 151 cities assessed for the 2024 LOSI